

LAST NAME: _____

Practice Quiz 10C — MATH 2055 — Fall 2026

Practice for Quiz 10

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

1. (a) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{p, q, s\}$, and define

$$f(1) = p, \quad f(2) = q, \quad f(3) = p, \quad f(4) = s, \quad f(5) = q.$$

For $X = \{2, 4\}$, compute $f(X)$ and $f^{-1}(f(X))$. Are the latter and X equal?

- (b) For real $r \neq 1$, state and prove by induction a closed formula for

$$1 + r + r^2 + \cdots + r^{n-1} \quad (n \geq 1).$$